

Alexander L. Li

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EDUCATION

Massachusetts Institute of Technology (MIT)

Exp. May 2028

Candidate for B.S. in Artificial Intelligence (6-4) & Mathematics (18)

Cambridge, MA

- **Relevant Coursework:** Deep Learning, Natural Language Processing, Computer Vision, Algorithms, Web Design Lab, C & Assembly, Real Analysis, Discrete Math, Linear Algebra, Differential Equations, Probability & Statistics
- **Activities:** Claude Builder Club (VP/Co-Founder), Logarithms A Cappella (Director), NCAA Men's Fencing

EXPERIENCE

Amazon Web Services (AWS)

Jun 2026 – Aug 2026

Software Development Engineer (Internship / On-Site)

Boston, MA

- In-Progress Summer 2026 Project — Snapshot Creation Team for Amazon Elastic Block Storage (EBS)
- Improving high-latency snapshot upload throughput by 900% w/ dynamic queue design & statistical methods
- Engineering C++ utility supporting latency tracking in a single pass, adoptable by other EBS server modules
- Validating correctness/performance across 6 latency profiles through unit, integration, & A/B benchmarking tests

Fetch.ai (Agentic AI Service)

Jun 2025 – Sep 2025

Junior Software Engineer (Internship / Remote)

Cambridge, UK

- Shipped 5+ AI agents to production with ~10k user interactions, facilitating dynamic service queries from an LLM
- Integrated multi-modal agent architecture by protocol-driven understanding & distributed agent orchestration
- Containerized agents with Docker, enabling edge-to-cloud communication by mailbox systems across agent nodes

Air Force Research Laboratory

May 2025 - Aug 2025

Deep Learning Researcher (Internship / On-site)

Wright-Patterson AFB, OH

- Trained a conditional diffusion model with PyTorch to reverse-engineer image generation parameters from Blender
- Designed multi-head attention U-Net conditioned on ResNet-encoded images to infer input parameter distributions
- Scaled training efficiency by 400% via multi-node distributed data parallel (DDP) & SLURM for batch scheduling
- Published as first author in SPIE Defense + Security Journal and presented work as a keynote speaker

PROJECTS

Seeded Diffusion Refinement for Music Transcription

May 2026

Python, PyTorch, NumPy, Diffusion Modeling, Computer Vision

- Established a diffusion-based piano audio-to-MIDI transcription pipeline; improving note-level onset F1 by 20%
- Engineered a ControlNet-style audio-conditioned denoiser & seeded DDIM sampler in PyTorch
- Designed mechanism ablations with 1,000-resample piece-level bootstrap CIs, isolating inference-time iteration.

Campus Marketplace Platform

Jul 2025

React.js, HTML, Tailwind CSS, Node.js, Express, FastAPI, PostgreSQL, Supabase, Docker, CLIP

- Built a scalable, mobile-first campus marketplace app with secure user auth & real-time item listings
- Implemented AI-powered category recommendations using OpenAI CLIP & FastAPI microservices
- Deployed in Dockerized containers with PostgreSQL & Supabase; architected for future cloud scaling

ChillDeck DJ (MIT Hackathon) - 2nd Place, Best UI Design

Jan 2025

React.js, HTML/CSS, TypeScript, RESTful APIs, Node.js, MongoDB, Figma

- Developed (as a team of 3) an interactive DJ platform allowing users to remix songs by isolating audio stems
- Engineered backend stem processing with Audioshake API & real-time audio waveforms via Wavesurfer.js
- Designed full user flow integrated with MongoDB: login, theming, uploads, & custom track playback

TECHNICAL SKILLS

Languages: Python, C/C++, Golang, TypeScript, JavaScript, Java, SQL, HTML/CSS

Frameworks/Technologies: React, Node.js, FastAPI, Flask, MongoDB, PostgreSQL, Docker, Git, Linux, PyTorch, TensorFlow, LangChain, LangGraph, SLURM, Apptainer/Singularity

Awards: USA Computing Olympiad Gold Division, 5th in Cryptography @ Science Olympiad Nationals